10/06/2025 Briana-Yves 1-1 Meeting Minutes

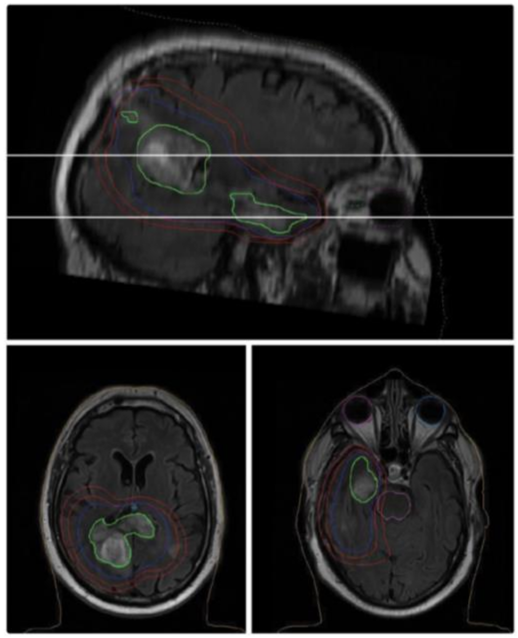
We discussed a new project: multifocal GBM detection. We will have 60 unifocal GBMs to fine-tune the nnUNET model and 43 multifocal GBMs. One-hot encoded mutations will be added as part of the classifier's features.

Background

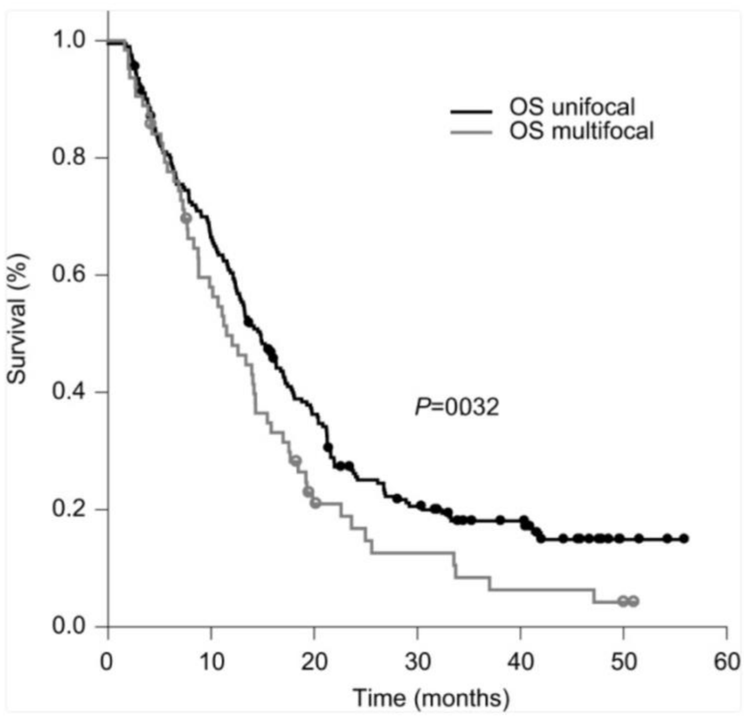
Glioblastoma (**GBM**) is the most common malignant primary tumor of the adult central nervous system, and despite aggressive multimodal therapies, with a median age at diagnosis of approximately 65 years, the median overall survival (OS) is 14.6–16.7 months [1, 2]. Most GBMs present as solitary (unifocal) lesions; however, up to 2–35 % [3], exhibit multifocal or multicentric dissemination (**mGBM**). Multifocal GBMs are characterized by multiple contrast-enhancing foci connected by white-matter or vascular pathways, whereas multicentric GBMs consist of spatially distinct, unconnected lesions. Clinically, patients with mGBM experience shorter survival and reduced surgical options compared with those with unifocal GBM (**uGBM**). MRI-particularly contrast-enhanced T1-weighted and T2-FLAIR sequences, remains the primary modality for mGBM detection. Imaging and histopathologic studies reveal that radiologically unifocal GBMs may harbor microscopic secondary foci that later coalesce or become visible on follow-up imaging [6]. These observations suggest that mGBMs often evolve from early unifocal stages through clonal migration and spatial diversification, rather than arising as independent lesions. Although prior machine-learning (ML) studies have stratified GBM patients using radiomic and prognostic features, no published AI/ML approaches have focused on predicting or detecting mGBM based on its correlation with imaging phenotypes. Early identification of mGBM could enhance clinical decision-making, treatment planning, and surveillance strategies.

Method

We will train a deep-learning segmentation model (nnU-Net) using the curated unifocal GBM data from the multi-institutional BraTS 2017 dataset, and fine-tune it on Johns Hopkins Hospital (JHH) data containing both unifocal and multifocal GBMs. Using the learned nnU-Net feature representations, we will then train a classifier on JHH unifocal GBMs with known clinical recurrence, balanced between unifocal and multifocal outcomes, to predict whether a unifocal GBM would recur as unifocal or multifocal (binary outcome). Finally, we will identify and visualize the nnU-Net feature kernels most influential to the classifier's ability to distinguish recurrence patterns.



Exemplary radiation treatment plan for a patient with multifocal GBM, 3D conformal radiotherapy to a dose of 60.0 Gy in 30 fractions.



Kaplan–Meier curve showing significantly worse OS of patients with mGBM as compared to uGBM (11.5 vs 14.8 months, P=0.032).

Reference

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